

# Legendre's Conjecture in Function Fields: the Classical Range, Full Monodromy, and the Open Range $q \leq d - 2$

Version v7 of *The Geometry of Prime Vacuums*

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Code and data: <https://github.com/Ruqing1963/legendre-function-field>

## Abstract

For a monic polynomial  $f \in \mathbb{F}_q[t]$  of degree  $d$ , the *Legendre interval*  $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$  is the function field analogue of  $[n^2, (n+1)^2]$ . We record the explicit form of a classical estimate (Hayes characters and Weil's theorem):  $|\sum_{P \in \mathcal{I}_f} \Lambda(P) - q^{d+1}| \leq (d-2)(q^d - q)$  in every characteristic. Together with a bound for prime powers in  $\mathcal{I}_f$ , it shows that  $\mathcal{I}_f$  contains an irreducible polynomial whenever  $q \geq d - 1$ . This supersedes all thresholds of earlier versions of this preprint, which overlooked it. The genuine analogue of Legendre's conjecture, with  $q$  fixed and  $d \rightarrow \infty$ , lies in the remaining range  $q \leq d - 2$ . We examine every Legendre interval for  $q = 2$  with  $d \leq 20$ , for  $q = 3$  with  $d \leq 10$ , for  $q = 4$  with  $d \leq 10$ , and for  $q = 5$  with  $d \leq 8$ . All of them contain irreducible polynomials, so the analogue of Legendre's conjecture holds for every  $q$  when  $d \leq 8$ . Over odd  $q$  the variance of the prime count across intervals is already close to the Keating–Rudnick prediction  $(d-2)q^{d+1}$ , proved only for  $q \rightarrow \infty$ , and the smallest normalized count approaches 1 as  $d$  grows. We state a conjecture with a heuristic based on these observations. We also give a self-contained proof that the geometric monodromy group of the family is  $S_{2d}$  for  $p > 2d$ , an exact twisted-variety identity, closed formulas for  $d \leq 3$ , and a list of corrections to v5 and v6.

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# 1 Introduction

## 1.1 The question

Legendre's conjecture asserts that there is a prime between  $n^2$  and  $(n+1)^2$  for every  $n \geq 1$ . It is open. The best unconditional result on prime gaps gives primes in  $[x, x + x^{0.525}]$  [1], and even the Riemann hypothesis only gives intervals of length about  $\sqrt{x} \log x$ .

In  $\mathbb{F}_q[t]$  put  $|g| = q^{\deg g}$ . For monic  $f$  of degree  $d$ , the polynomials  $g$  with  $|g - f^2| \leq |f|$  form the *Legendre interval*

$$\mathcal{I}_f = \{f^2 + s : s \in \mathbb{F}_q[t], \deg s \leq d\}, \quad |\mathcal{I}_f| = q^{d+1}.$$

Its elements are the monic polynomials of degree  $2d$  whose coefficients of  $t^{2d-1}, \dots, t^{d+1}$  agree with those of  $f^2$ . We write

$$N_{\text{irr}}(f) = \#\{P \in \mathcal{I}_f : P \text{ irreducible}\}, \quad \Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P),$$

where  $\Lambda(P) = \deg \pi$  if  $P = \pi^r$  with  $\pi$  irreducible and  $\Lambda(P) = 0$  otherwise. Then  $2d N_{\text{irr}}(f) = \Psi(f) - E(f)$ , where  $E(f)$  collects the prime powers  $\pi^r$ ,  $r \geq 2$ . The prime polynomial theorem predicts  $N_{\text{irr}}(f) \approx q^{d+1}/(2d)$ . *Does every Legendre interval contain an irreducible polynomial?*

## 1.2 What is known, and what this paper adds

**The classical range**  $q \geq d - 1$ . The interval  $\mathcal{I}_f$  is a class of polynomials with  $\ell = d - 1$  prescribed leading coefficients. Such classes are the fibres of Hayes's characters [9], whose  $L$ -functions are polynomials of degree at most  $\ell - 1$  satisfying the Riemann hypothesis by Weil's theorem. This gives explicit bounds for irreducible polynomials with prescribed leading coefficients (Hsu [10], Cohen [4, Thm. 2.1]; see [7, Cors. 1–2]). For Legendre intervals they read as follows.

**Theorem 1** (Section 2). *For every prime power  $q$  and every monic  $f$  of degree  $d \geq 2$ ,*

$$|\Psi(f) - q^{d+1}| \leq (d-2)(q^d - q).$$

*Consequently  $N_{\text{irr}}(f) \geq 1$  whenever  $q \geq d - 1$ , in every characteristic.*

The first statement is classical in substance; we give a short proof to fix constants. The consequence for  $q \in \{d-1, d\}$  uses an elementary bound on  $E(f)$  (Lemma 2.3). At  $q = d-2$  the estimate gives nothing, because the main term  $q^{d+1}$  and the error  $(d-2)q^d$  are then equal. This is the function field form of the familiar fact that the Riemann hypothesis *just* fails to reach Legendre’s conjecture over  $\mathbb{Z}$ .

Earlier versions of this preprint missed Theorem 1. Version v5 claimed the threshold  $q > (8d)^{2d+6}$  without valid proof, and version v6 proved thresholds such as  $q > 1.65 \times 10^4$  for  $d = 4$ . Both are far weaker than  $q \geq d-1$ ; see Appendix B. The asymptotic  $N_{\text{irr}} = q^{d+1}/2d + O_d(q^{d+1/2})$  for fixed  $d$ , the “main theorem” of v5, is contained in Theorem 1 and also in the short-interval theorem of Bank, Bary-Soroker and Rosenzweig [2]. Their result, and Sawin’s sharper bound [19], concern shorter intervals or larger  $q$ ; at the Legendre length the classical estimate is the relevant one.

**The open range  $q \leq d-2$ .** Here no general method is known. Reversing polynomials turns  $\mathcal{I}_f$  into a residue class modulo  $t^d$  among polynomials of degree  $2d$ , at level of distribution exactly  $1/2$ . Sawin and Shusterman [20] pass level  $1/2$  for squarefree moduli when  $q$  is large compared with  $p$ , but the moduli  $t^d$  are not covered. Results on prescribed coefficients [18, 8] allow far fewer than the  $d-1$  coefficients needed here. We therefore study this range by exhaustive computation.

**Theorem 2** (Section 3). *Every Legendre interval over  $\mathbb{F}_q$  contains an irreducible polynomial when  $q = 2, d \leq 20$ ;  $q = 3, d \leq 10$ ;  $q = 4, d \leq 10$ ;  $q = 5, d \leq 8$ . Together with Theorem 1, the function field analogue of Legendre’s conjecture holds for every prime power  $q$  when  $d \leq 8$ .*

The computations also measure how the prime count fluctuates from interval to interval. For odd  $q$  the Legendre intervals are *all* the intervals of length  $q^{d+1}$  in degree  $2d$ , and Keating and Rudnick [13] proved that the variance of  $\Psi$  over them is  $\sim (d-2)q^{d+1}$  as  $q \rightarrow \infty$ . We find this already accurate for  $q = 3$  and  $q = 5$  (Table 1). A Gaussian model with this variance predicts that the smallest normalized count tends to 1 as  $d \rightarrow \infty$ , which is what the data show (Figure 1). This leads to:

**Conjecture** (Section 4). *For every prime power  $q$  and every  $d \geq 1$ , every Legendre interval in  $\mathbb{F}_q[t]$  contains an irreducible polynomial. Moreover, for fixed  $q$ ,  $\min_f \Psi(f)/q^{d+1} \rightarrow 1$  as  $d \rightarrow \infty$ .*

**Other results kept from v6.** For  $p > 2d$  we give a self-contained proof that the geometric monodromy group of  $f^2 + \sum_{i \leq d} a_i t^i$  is  $S_{2d}$ , for every monic  $f$  (Theorem 5.7). We also prove an exact identity  $\Psi(f) = \#W(\mathbb{F}_q)$  for an explicit twisted root variety  $W$  (Theorem 6.2), and closed formulas for  $N_{\text{irr}}$  when  $d \leq 3$  (Section 7). These are not needed for Theorems 1 and 2, but they explain the factorization statistics in Figure 3 and give independent checks of the code.

### 1.3 About this revision

Appendix A lists the errors of v5, and Appendix B the omissions of v6 corrected here. The main changes from v6 are Theorem 1, which replaces all of v6’s explicit thresholds, and the new computations of Section 3.

## 2 The classical range

### 2.1 Hayes classes

For monic  $g$  of degree  $k$  put  $\tilde{g}(u) = u^k g(1/u) \in \mathbb{F}_q[u]$ ; its constant term is 1. Let  $\ell \geq 0$  and let  $\mathcal{G}_\ell$  be the group of power series  $1 + c_1 u + \dots$  modulo  $u^{\ell+1}$ , of order  $q^\ell$ . The map

$g \mapsto [g] := \tilde{g} \bmod u^{\ell+1}$  is multiplicative, and  $[g]$  records exactly the  $\ell$  coefficients of  $g$  below the leading one. With  $\ell = d - 1$ ,  $\mathcal{I}_f$  is the set of monic  $P$  of degree  $2d$  with  $[P] = [f^2]$ .

**Lemma 2.1.** *Let  $\chi \neq 1$  be a character of  $\mathcal{G}_\ell$  and  $\chi(g) = \chi([g])$  for monic  $g$ . Then  $L(z, \chi) = \sum_{g \text{ monic}} \chi(g) z^{\deg g}$  is a polynomial of degree at most  $\ell - 1$ , and its inverse roots have absolute value  $q^{1/2}$  or 1.*

*Proof.* For  $k \geq \ell$  the map  $g \mapsto [g]$  from monic polynomials of degree  $k$  onto  $\mathcal{G}_\ell$  has all fibres of size  $q^{k-\ell}$ , so  $\sum_{\deg g=k} \chi(g) = 0$ . Hence  $L$  is a polynomial of degree  $\leq \ell - 1$ . These are  $L$ -functions of characters of a ray class group of  $\mathbb{F}_q(t)$ , with conductor supported at the place  $t = \infty$ , and the Riemann hypothesis for them is Weil's theorem; see [9] and [13, §4], where they appear as even Dirichlet characters modulo  $u^{\ell+1}$  after the substitution  $t \mapsto 1/u$ .  $\square$

**Theorem 2.2** (Theorem 1, first part). *For every  $q$  and every monic  $f$  of degree  $d \geq 2$ ,  $|\Psi(f) - q^{d+1}| \leq (d-2)(q^d - q)$ . For  $d = 2$  this means  $\Psi(f) = q^3$  exactly.*

*Proof.* Let  $\ell = d - 1$  and  $n = 2d$ . By orthogonality,  $\Psi(f) = q^{-\ell} \sum_{\chi} \bar{\chi}([f^2]) \sum_{\deg P=n} \Lambda(P) \chi(P)$ . The trivial character contributes  $q^{-\ell} \cdot q^n = q^{d+1}$ . For  $\chi \neq 1$  write  $L(z, \chi) = \prod_i (1 - \omega_i z)$  with at most  $\ell - 1$  factors. Taking the logarithmic derivative gives  $\sum_{\deg P=n} \Lambda(P) \chi(P) = -\sum_i \omega_i^n$ , of absolute value at most  $(\ell - 1)q^{n/2} = (d-2)q^d$ . There are  $q^\ell - 1$  nontrivial characters, each weighted by  $q^{-\ell}$ , so  $|\Psi(f) - q^{d+1}| \leq (1 - q^{-(d-1)})(d-2)q^d = (d-2)(q^d - q)$ .  $\square$

**Lemma 2.3** (Prime powers in a Legendre interval). *Let  $E(f) = \Psi(f) - 2d N_{\text{irr}}(f)$ . Then*

$$E(f) \leq dq^a + 2q^{\lfloor 2d/3 \rfloor}, \quad a = \begin{cases} 1, & p \text{ odd}, \\ \lceil (d+1)/2 \rceil, & p = 2. \end{cases}$$

*Moreover  $E(f) \leq \sum_{e|2d, e < 2d} q^e < q^d + 2q^{\lfloor 2d/3 \rfloor}$  in all cases.*

*Proof.*  $E(f)$  is the sum of  $\deg \pi$  over irreducible  $\pi$  of degree  $e = 2d/r$ ,  $r \geq 2$ , with  $\pi^r \in \mathcal{I}_f$ . Since  $e \cdot \#\{\pi : \deg \pi = e\} \leq q^e$ , the terms with  $r \geq 3$  contribute at most  $\sum_{e \leq 2d/3} q^e < 2q^{\lfloor 2d/3 \rfloor}$ , and all terms together at most  $\sum_{e|2d, e < 2d} q^e$ . For  $r = 2$ , write  $\pi = t^d + \pi_{d-1}t^{d-1} + \dots + \pi_0$ . If  $p$  is odd, the coefficient of  $t^{2d-j}$  in  $\pi^2$  is  $2\pi_{d-j} + \Phi_j(\pi_{d-1}, \dots, \pi_{d-j+1})$  for a universal polynomial  $\Phi_j$ , so the  $d-1$  prescribed coefficients determine  $\pi_{d-1}, \dots, \pi_1$ , and at most  $q$  polynomials  $\pi$  qualify. If  $p = 2$ , then  $\pi^2 = \sum_i \pi_i^2 t^{2i}$ , the prescribed coefficients determine  $\pi_{d-1}, \dots, \pi_{d-\lfloor (d-1)/2 \rfloor}$ , and at most  $q^{\lceil (d+1)/2 \rceil}$  polynomials qualify. Each contributes  $d$ .  $\square$

**Corollary 2.4** (Theorem 1, second part). *If  $q \geq d - 1$ , then  $N_{\text{irr}}(f) \geq 1$  for every monic  $f$  of degree  $d$ .*

*Proof.* For  $d \leq 2$  see Theorem 7.1. Let  $d \geq 3$ . By Theorem 2.2,  $2d N_{\text{irr}}(f) \geq q^{d+1} - (d-2)(q^d - q) - E(f) = q^d(q - d + 2) + (d-2)q - E(f)$ .

If  $q \geq d$  the right side is at least  $2q^d + (d-2)q - E(f)$ , which is positive because  $E(f) < q^d + 2q^{\lfloor 2d/3 \rfloor} \leq 2q^d$  (note  $\lfloor 2d/3 \rfloor \leq d-1$  and  $q \geq 2$ ).

If  $q = d - 1$  the right side is  $q^d + (d-2)q - E(f)$ . For odd  $q$  (so  $d \geq 4$ ), Lemma 2.3 gives  $E(f) \leq dq + 2q^{d-2}$ , since  $\lfloor 2d/3 \rfloor \leq d-2$  for  $d \geq 4$ ; and  $2q + 2q^{d-2} < q^d$  because  $q \geq 3$ . For even  $q$  we have  $d = q + 1$  odd and  $E(f) \leq dq^{(d+1)/2} + 2q^{\lfloor 2d/3 \rfloor}$ , which is less than  $q^d$  for  $d \geq 5$ ,  $q \geq 4$ . The single remaining case  $(q, d) = (2, 3)$  is checked directly:  $N_{\text{irr}} = 2$  for every  $f$  (Table 1).  $\square$

*Remark 2.5.* The second part of Theorem 1 is sharp for this method: when  $q = d - 2$  the lower bound  $q^d(q - d + 2) + (d-2)q - E(f)$  equals  $q^2 - E(f)$ , which is negative. Theorem 1 also gives  $N_{\text{irr}} = q^{d+1}/2d + O(dq^d)$  for fixed  $d$ , with an explicit constant. This is stronger than the error  $O_d(q^{d+1/2})$  of [2]. Sawin's bound [19] has the better  $q$ -exponent  $q^{d/2+1}$  but a constant of size  $(2d+2)^{3d}$ ; it is the right tool for  $q$  very large compared with  $d$ , not for existence.

All 174 exact counts with  $q > d$  computed for v6 satisfy the bounds of Theorem 2.2 (data/interval\_counts.csv).

### 3 The open range $q \leq d - 2$

#### 3.1 Method

By Theorem 1 only the pairs with  $q \leq d - 2$  need checking. The substitutions  $t \mapsto \lambda t + b$  ( $\lambda \in \mathbb{F}_q^\times$ ,  $b \in \mathbb{F}_q$ ) preserve irreducibility and map Legendre intervals to Legendre intervals:  $f$  goes to  $\lambda^{-d} f(\lambda t + b)$ . We list all distinct Legendre intervals, group them into orbits under these substitutions, and for one representative of each orbit we test every element for irreducibility. We also compute  $E(f)$  by enumerating the candidate prime powers, so  $\Psi(f) = 2d N_{\text{irr}}(f) + E(f)$  is known exactly.

Field arithmetic uses addition and multiplication tables for  $\mathbb{F}_q$ , so prime and non-prime  $q$  are handled by the same code. Irreducibility is decided by Ben-Or's test: a monic  $P$  of degree  $n$  is irreducible if and only if  $\gcd(t^{q^i} - t, P) = 1$  for  $1 \leq i \leq n/2$ . The implementation is cross-checked against an independent distinct-degree factorization for prime  $q$ , against a pure-Python implementation for  $q = 4, 8, 9$ , and against the twisted-variety count of Section 6. A further check is exact: for odd  $q$  every residue class of leading coefficients is a Legendre interval, so the orbit-weighted sum of  $\Psi$  must equal  $q^{d-1} \cdot q^{d+1}$ . It does in every case tested.

In characteristic 2 the Legendre intervals form a thin subfamily. Indeed  $f^2 = \sum c_i^2 t^{2i}$ , so every prescribed coefficient in an odd position is 0. There are only  $q^{\lfloor (d-1)/2 \rfloor}$  Legendre intervals, which is why  $q = 2$  can be pushed much further than  $q = 3$ .

#### 3.2 Results

**Theorem 3.1** (Theorem 2). *Every Legendre interval over  $\mathbb{F}_q$  contains an irreducible polynomial for  $q = 2$  and  $d \leq 20$ ,  $q = 3$  and  $d \leq 10$ ,  $q = 4$  and  $d \leq 10$ , and  $q = 5$  and  $d \leq 8$ . Consequently, for every prime power  $q$  and every  $d \leq 8$ , every Legendre interval contains an irreducible polynomial.*

*Proof.* The first statement is the computation summarized in Table 1. For the second, let  $d \leq 8$ . If  $q \geq d - 1$  apply Theorem 1. Otherwise  $q \leq d - 2 \leq 8 - 2$ , so  $q$  is one of the prime powers listed, and the first statement applies.  $\square$

#### 3.3 Observations

- (i) No empty Legendre interval occurs. The smallest normalized count in the open range is 0.375, at  $(q, d) = (2, 6)$ .
- (ii) For odd  $q$  the empirical variance of  $\Psi$  approaches  $(d - 2)q^{d+1}$ , although Keating and Rudnick prove the asymptotic only for  $q \rightarrow \infty$ . For  $q = 3$  the relative deviation falls from 21% at  $d = 6$  to 1.4% at  $d = 10$ ; for  $q = 5$  it is at most 11% for  $5 \leq d \leq 8$ .
- (iii) In characteristic 2 the Legendre intervals are a special subfamily and the variance fluctuates more. Small cases also show a downward bias in  $N_{\text{irr}}$ , for example  $\min_f 2d N_{\text{irr}}/q^{d+1} = 0.375$  at  $(q, d) = (2, 6)$ . The bias fades as  $d$  grows.

### 4 A conjecture

**Conjecture 4.1.** *For every prime power  $q$  and every  $d \geq 1$ , every Legendre interval in  $\mathbb{F}_q[t]$  contains an irreducible polynomial. Moreover  $\min_f \Psi(f)/q^{d+1} \rightarrow 1$  as  $d \rightarrow \infty$  with  $q$  fixed.*

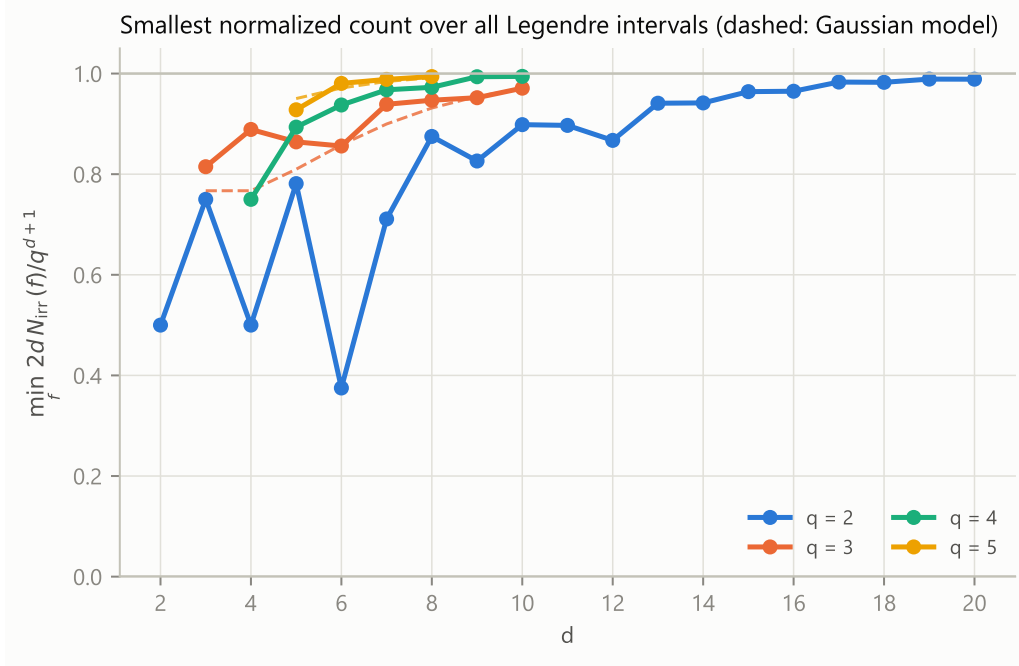


Figure 1: The smallest normalized count  $\min_f 2d N_{\text{irr}}(f)/q^{d+1}$  in the open range. It never reaches 0 and approaches 1 as  $d$  grows.

**Heuristic.** Model the values  $\Psi(f)$  over the  $q^{d-1}$  Legendre intervals ( $q$  odd) as Gaussian with mean  $q^{d+1}$  and variance  $(d-2)q^{d+1}$ , as suggested by [13] and by Table 1. The minimum of  $N$  such variables is about  $-\sqrt{2 \log N}$  standard deviations, so

$$\min_f \frac{\Psi(f)}{q^{d+1}} \approx 1 - \sqrt{\frac{2(d-1)(d-2) \log q}{q^{d+1}}} \rightarrow 1.$$

The deficit is at most 0.51 over all  $q \geq 2$  and  $d \geq 3$ , attained at  $(q, d) = (2, 5)$ , so the model predicts no empty interval at all. Since  $E(f)$  is at most of order  $q^{\lfloor 2d/3 \rfloor}$  for odd  $q$ , the same holds for  $N_{\text{irr}}$ . The observed minima are consistent with the prediction (Figure 1). The heuristic is not a proof. The Gaussian model is only known in the limit  $q \rightarrow \infty$ , and the extreme tail of the distribution is exactly what a proof would have to control.

## 5 The monodromy group

This section and the next two are kept from v6, with minor changes. Write  $\mathcal{M} = \mathbb{A}^{d+1}$  with coordinates  $\mathbf{a} = (a_0, \dots, a_d)$  and  $F(t, \mathbf{a}) = f(t)^2 + \sum_{i \leq d} a_i t^i$ .

### 5.1 Preliminaries

**Lemma 5.1** (Membership in  $\mathcal{I}_f$ ). *A monic  $P \in \mathbb{F}_q[t]$  of degree  $2d$  lies in  $\mathcal{I}_f$  if and only if its coefficients of  $t^{2d-1}, \dots, t^{d+1}$  agree with those of  $f^2$ . If  $p > d-1$ , this holds if and only if the power sums  $p_j(P) = \sum_{P(\rho)=0} \rho^j$  satisfy  $p_j(P) = 2p_j(f)$  for  $1 \leq j \leq d-1$ .*

*Proof.* Newton's identities express the power sums  $p_1, \dots, p_{d-1}$  through the top  $d-1$  coefficients, and conversely when  $1, \dots, d-1$  are invertible. Finally  $p_j(f^2) = 2p_j(f)$ .  $\square$

**Lemma 5.2** (Inertia at a simple node). *Let  $R$  be a complete discrete valuation ring with algebraically closed residue field of characteristic  $\neq 2$ , fraction field  $K$  and valuation  $v$ . Let  $P \in R[t]$*

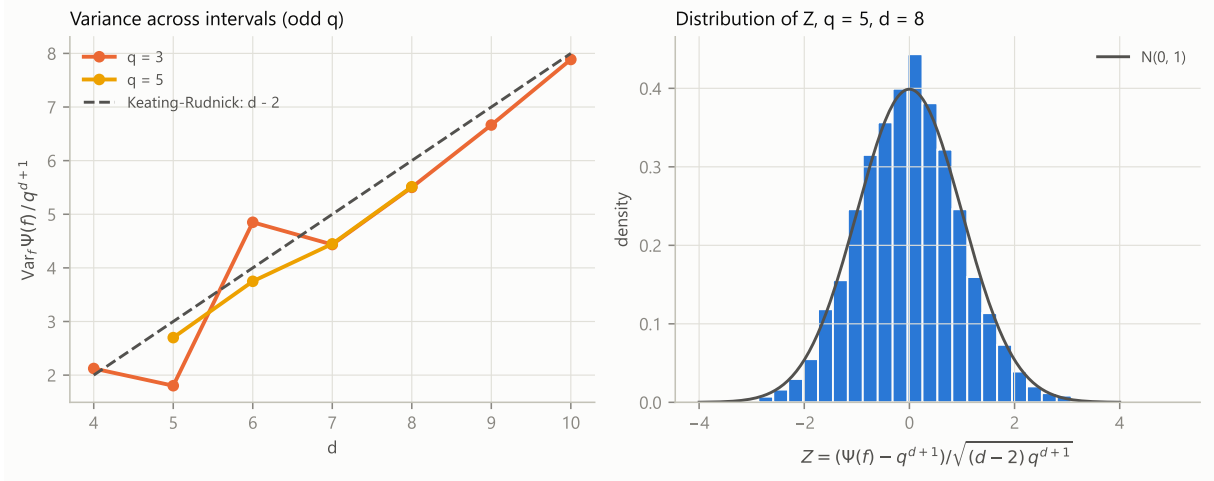


Figure 2: Left:  $\text{Var}_f \Psi(f)/q^{d+1}$  over all Legendre intervals for odd  $q$ , against the Keating–Rudnick limit  $d-2$ . Right: the distribution of  $Z = (\Psi(f) - q^{d+1})/\sqrt{(d-2)q^{d+1}}$  for the largest odd case, against the standard normal density.

be monic of degree  $n$  whose reduction  $\bar{P}$  has exactly one double root  $\beta$  and  $n-2$  simple roots, and suppose  $v(\text{disc}P) = 1$ . Then the splitting field of  $P$  over  $K$  is a quadratic extension whose Galois group acts on the roots of  $P$  as a transposition.

*Proof.* Write  $\bar{P} = (t - \beta)^2 \bar{L}$  with  $\bar{L}(\beta) \neq 0$  and  $\bar{L}$  squarefree. By Hensel’s lemma  $P = QL$  with  $Q, L \in R[t]$  monic lifting  $(t - \beta)^2$  and  $\bar{L}$ , and  $L$  splits into linear factors over  $R$ . Since  $\text{disc}P = \text{disc}Q \cdot \text{disc}L \cdot \text{Res}(Q, L)^2$  and both  $\text{disc}L$  and  $\text{Res}(Q, L)$  are units,  $v(\text{disc}Q) = 1$ . An element of odd valuation is not a square in  $K$ , so  $Q$  is irreducible over  $K$ , and the nontrivial automorphism of its splitting field swaps the two roots of  $Q$  and fixes the roots of  $L$ .  $\square$

**Lemma 5.3** (Primitivity and decomposition; cf. [21, 17]). *Let  $K$  be a field,  $G \in K[t]$  of degree  $n \geq 2$  with  $G' \neq 0$ , and  $x$  transcendental over  $K$ . If  $\text{Gal}(G(t) - x / K(x))$ , acting on the roots of  $G(t) - x$ , is imprimitive, then  $G = g \circ h$  with  $g, h \in K[t]$  and  $\deg g, \deg h \geq 2$ .*

*Proof.* Let  $\tau$  be a root. Blocks containing  $\tau$  correspond to fields  $K(x) \subseteq L \subseteq K(\tau)$ , so imprimitivity gives a proper intermediate field  $L$ , and  $L = K(h)$  by Lüroth’s theorem. Since  $x = G(\tau)$  is a polynomial in  $\tau$ , the place  $\tau = \infty$  is the only place of  $K(\tau)$  above  $x = \infty$ , and it is  $K$ -rational; so is its restriction to  $L$ . After a Möbius transformation over  $K$  we may take that place to be  $h = \infty$ . Then  $h$  has its only pole at  $\tau = \infty$ , so  $h \in K[\tau]$ , and likewise  $x = g(h)$  with  $g \in K[X]$ . Both degrees are  $\geq 2$  because  $L$  is a proper intermediate field.  $\square$

## 5.2 Full monodromy

Let  $\mathcal{D} \subset \mathcal{M}$  be the zero locus of  $\text{disc}_t F$  and  $\mathcal{M}^\circ = \mathcal{M} \setminus \mathcal{D}$ . The geometric monodromy group of the finite étale cover  $\{F = 0\} \rightarrow \mathcal{M}^\circ$  is  $G_{\text{geom}} = \text{Gal}(F/\mathbb{F}_q(\mathbf{a}))$ . Put  $K' = \mathbb{F}_q(a_1, \dots, a_d)$  and  $G(t) = f(t)^2 + a_1 t + \dots + a_d t^d$ , so  $F = G(t) + a_0$  and  $G_{\text{geom}} = \text{Gal}(G(t) - x/K'(x))$  with  $x = -a_0$ .

**Proposition 5.4** (Transitivity).  *$F$  is irreducible over  $\mathbb{F}_q(\mathbf{a})$ , so  $G_{\text{geom}}$  is transitive.*

*Proof.*  $F$  has degree one in  $a_0$  with coprime coefficients 1 and  $f^2 + \sum_{i \geq 1} a_i t^i$ , so it is irreducible in  $\mathbb{F}_q[\mathbf{a}, t]$ , hence in  $\mathbb{F}_q(\mathbf{a})[t]$  by Gauss’s lemma.  $\square$

**Proposition 5.5** (Primitivity). *If  $p > d$ , then  $G_{\text{geom}}$  is primitive.*



*Proof.* For  $d = 1$  this is automatic. Let  $d \geq 2$ ; note  $G' \neq 0$  because of the term  $a_1 t$ . If  $G_{\text{geom}}$  were imprimitive, Lemma 5.3 would give  $G = g \circ h$  over  $K'$  with  $\deg h = m$ ,  $\deg g = n$ ,  $mn = 2d$ ,  $m, n \geq 2$ , so  $m \leq d$ . Normalize  $h$  to be monic with  $h(0) = 0$ .

*Step 1.* For  $1 \leq j \leq m - 1$  we have  $2d - j \geq d + 1$ , so the coefficient of  $t^{2d-j}$  in  $G$  is that of  $f^2$ , an element of  $\mathbb{F}_q$ . Also  $g(h) = h^n + (\text{terms of degree} \leq 2d - m)$  and  $[t^{2d-j}]h^n = n h_{m-j} + \Phi_j(h_{m-1}, \dots, h_{m-j+1})$  with  $\Phi_j$  universal. As  $n \leq d < p$ , induction gives  $h \in \mathbb{F}_q[t]$ .

*Step 2.* Let  $U = \text{span}_{\mathbb{F}_q}\{1, h, \dots, h^n\}$ . Then  $G \in U \otimes K'$ . For any linear form  $\ell$  vanishing on  $U$ ,  $0 = \ell(f^2) + \sum_{i=1}^d a_i \ell(t^i)$ , so  $\ell(t) = 0$  because the  $a_i$  are algebraically independent. Hence  $t \in U$ , impossible since nonconstant elements of  $U$  have degree divisible by  $m \geq 2$ .  $\square$

**Proposition 5.6** (A transposition). *If  $p > 2d$ , then  $G_{\text{geom}}$  contains a transposition.*

*Proof. Step 1: a fibre with exactly one double root.* If  $d = 1$  take  $P^* = (t - \beta)^2$ . For  $d \geq 2$  fix  $\beta \in \overline{\mathbb{F}}_q$ , divide  $f^2 = (t - \beta)^2 Q + r$  with  $\deg r \leq 1$ , and put  $P^* = (t - \beta)^2(Q + c)$ . Then  $\deg(P^* - f^2) \leq 2 \leq d$ , so  $P^*$  is a fibre of  $F$ . Since  $Q'$  has leading coefficient  $2d - 2 \neq 0$  in  $\mathbb{F}_q$ , all but finitely many  $c$  make  $R = Q + c$  squarefree with  $R(\beta) \neq 0$ .

*Step 2: the discriminant vanishes to order one along  $\varepsilon \mapsto \mathbf{a}^* + \varepsilon e_0$ .* As  $\partial_t F = P^{*'} is independent of  $\varepsilon$  with leading coefficient  $2d \neq 0$ ,  $\text{disc}_t(P^* + \varepsilon) = u \prod_{P^{*'}(\gamma)=0} (P^*(\gamma) + \varepsilon)$  with  $u \neq 0$ . Now  $P^{*'} = (t - \beta)(2R + (t - \beta)R')$  has  $\beta$  as a simple root, and no root of  $R$  is a root of  $P^{*'}$ . So the order of vanishing at  $\varepsilon = 0$  is exactly 1.$

*Step 3.* The inertia group at  $\varepsilon = 0$  lies in  $G_{\text{geom}}$  and is generated by a transposition by Lemma 5.2.  $\square$

**Theorem 5.7.** *If  $p > 2d$ , then  $G_{\text{geom}} = S_{2d}$ , and also  $\text{Gal}(F/\mathbb{F}_q(\mathbf{a})) = S_{2d}$ .*

*Proof.* A primitive permutation group containing a transposition is the full symmetric group [5, Thm. 3.3A]. The arithmetic group contains  $G_{\text{geom}}$  and lies in  $S_{2d}$ .  $\square$

*Remark 5.8.* No squarefree hypothesis on  $f$  is needed. By [2, Prop. 3.6],  $G_{\text{geom}} = S_{2d}$  also holds for every  $p$  when  $d \geq 3$ , and for odd  $p$  when  $d = 2$ .

## 6 The twisted root variety

Let  $e_j$  be the elementary symmetric polynomials in  $u = (u_1, \dots, u_{2d})$  and  $\gamma_j = (-1)^j [t^{2d-j}]f^2$ . The root variety is  $V = \{u \in \mathbb{A}^{2d} : e_j(u) = \gamma_j, 1 \leq j \leq d-1\}$ . It is the variety  $X_{n,m,\mathbf{c}}$  of [19] with  $n = 2d$ ,  $m = d-1$ .

**Proposition 6.1.**  *$V$  is a reduced complete intersection of pure dimension  $d+1$  and degree at most  $(d-1)!$ , and  $V_{\overline{\mathbb{F}}_q}$  is irreducible if and only if  $G_{\text{geom}} = S_{2d}$ .*

*Proof.* Every component has dimension  $\geq d+1$  by Krull's theorem and  $\leq d+1$  because the map  $u \mapsto \prod(t - u_i)$  is finite; the degree bound is the refined Bézout inequality [6, Ex. 8.4.6]. The map is étale where the  $u_i$  are distinct, so  $V$  is generically reduced, hence reduced. Over  $\mathcal{M}^\circ$ ,  $V$  is an  $S_{2d}$ -torsor, and its number of geometric components is  $[S_{2d} : G_{\text{geom}}]$ .  $\square$

Fix an  $\mathbb{F}_q$ -basis  $\omega_1, \dots, \omega_{2d}$  of  $\mathbb{F}_{q^{2d}}$  with Moore matrix  $M = (\omega_i^{q^{k-1}})_{k,i}$ , and let  $W \subset \mathbb{A}^{2d}$  be defined by  $e_j(Mx) = \gamma_j$ . Frobenius permutes the linear forms  $(Mx)_k$  cyclically, so  $W$  is defined over  $\mathbb{F}_q$ ; it is isomorphic to  $V$  over  $\mathbb{F}_{q^{2d}}$ , and  $x \mapsto \sum x_i \omega_i$  identifies  $W(\mathbb{F}_q)$  with  $\{\beta \in \mathbb{F}_{q^{2d}} : \chi_\beta \in \mathcal{I}_f\}$ , where  $\chi_\beta = \prod_k (t - \beta^{q^k})$ .

**Theorem 6.2.** *For every  $q$  and  $f$ :  $\#W(\mathbb{F}_q) = \Psi(f)$ , and  $E(f)$  is the number of  $\beta$  in proper subfields of  $\mathbb{F}_{q^{2d}}$  with  $\chi_\beta \in \mathcal{I}_f$ . Hence  $2d N_{\text{irr}}(f) = \#W(\mathbb{F}_q) - E(f)$ .*



*Proof.* If  $\chi_\beta \in \mathcal{I}_f$  then  $\chi_\beta = \pi^r$  with  $\pi$  the minimal polynomial of  $\beta$ , and each  $\pi$  arises from exactly  $\deg \pi$  elements  $\beta$ . Irreducible  $P \in \mathcal{I}_f$  correspond to  $r = 1$ .  $\square$

*Remark 6.3.* Theorem 6.2 turns  $\Psi(f)$  into a point count on an explicit variety. With the Lang–Weil bound of Cafure and Matera [3] it gave the thresholds of v6, which Theorem 1 now supersedes (Appendix B). The identity remains useful for exact formulas and as an independent test of the code.

## 7 Exact formulas for $d \leq 3$

**Theorem 7.1.** (i) If  $d = 1$  then  $N_{\text{irr}} = (q^2 - q)/2$ .

(ii) If  $d = 2$  then  $N_{\text{irr}} = (q^3 - q)/4$  for odd  $q$  and  $N_{\text{irr}} = (q^3 - q^2)/4$  for even  $q$ .

(iii) If  $d = 3$ ,  $p \geq 5$ ,  $f = t^3 + c_2 t^2 + c_1 t + c_0$ ,  $\Delta = c_2^2 - 3c_1$  and  $\eta$  is the quadratic character of  $\mathbb{F}_q$ , then

$$6 N_{\text{irr}} = \begin{cases} q^4 - \eta(2\Delta) q^2 - q + \eta(-3) + \eta(2\Delta) - 1, & \Delta \neq 0, \\ q^4 - q - (q - 1) \eta(-3), & \Delta = 0. \end{cases}$$

*Proof.* (i)  $\mathcal{I}_f$  is the set of all monic quadratics. (ii) By Theorem 2.2,  $\Psi(f) = q^3$ . The prime powers in  $\mathcal{I}_f$  are the squares  $\pi^2$  with  $\deg \pi = 2$  or  $\pi = (t - b)$ ; by Theorem 6.2,  $E(f)$  is the number of  $\beta \in \mathbb{F}_{q^2}$  with  $\chi_\beta \in \mathcal{I}_f$ . For such  $\beta$ ,  $\chi_\beta = (t^2 - \text{Tr} \beta t + N\beta)^2$  has  $t^3$ -coefficient  $-2\text{Tr} \beta$ , which must equal  $[t^3]f^2 = 2c_1$ . For odd  $q$  this has  $q$  solutions; for even  $q$  both sides vanish and all  $q^2$  elements qualify. (iii) This is the quadric count of v6, Theorem 6.2:  $W$  is a quadric in a hyperplane of  $\mathbb{A}^6$ , and the counts for nondegenerate quadratic forms [15, Thms. 6.26, 6.27] give  $\#W(\mathbb{F}_q)$  and  $E(f)$  exactly. It agrees with Kuz'min's formula for two prescribed coefficients [14, 16].  $\square$

All 141 applicable exact counts of `data/interval_counts.csv` agree with (i)–(iii), and (ii) for even  $q$  was checked for  $q = 2, 4, 8$ .

## 8 Discussion

**Where the difficulty lies.** Theorem 1 shows that the analogue of Legendre's conjecture is a consequence of the Riemann hypothesis for Hayes  $L$ -functions as long as  $q \geq d - 1$ . Below that, the  $d - 2$  inverse roots of each of the  $q^{d-1} - 1$   $L$ -functions could in principle conspire, and nothing rules this out. Over the integers the corresponding statement is that the Riemann hypothesis gives primes in  $[x, x + c\sqrt{x} \log x]$  but not in  $[x, x + 2\sqrt{x}]$ . In both settings the missing input is cancellation among zeros, not the location of zeros.

**What the data suggest.** The variance across intervals matches the random-matrix prediction of Keating and Rudnick even for  $q = 3$ , so the zeros of the relevant  $L$ -functions do behave independently in the range we can compute. If that persists, empty intervals never occur (Section 4).

### Open problems.

- (1) Prove Conjecture 4.1 for one fixed  $q$ , for instance  $q = 2$ , where the Legendre intervals form a thin and structured family.
- (2) Prove the Keating–Rudnick variance for fixed  $q$  and  $d \rightarrow \infty$ , even in the Legendre range.
- (3) Extend Theorem 2 to  $d = 9$ . This requires  $q = 5$  and  $q = 7$  with  $d = 9$  in full, about  $2 \times 10^{11}$  and  $4 \times 10^{13}$  irreducibility tests with the present method.

Factorization types in  $\mathcal{I}_{t^d}$  follow the cycle types of  $S_{2d}$

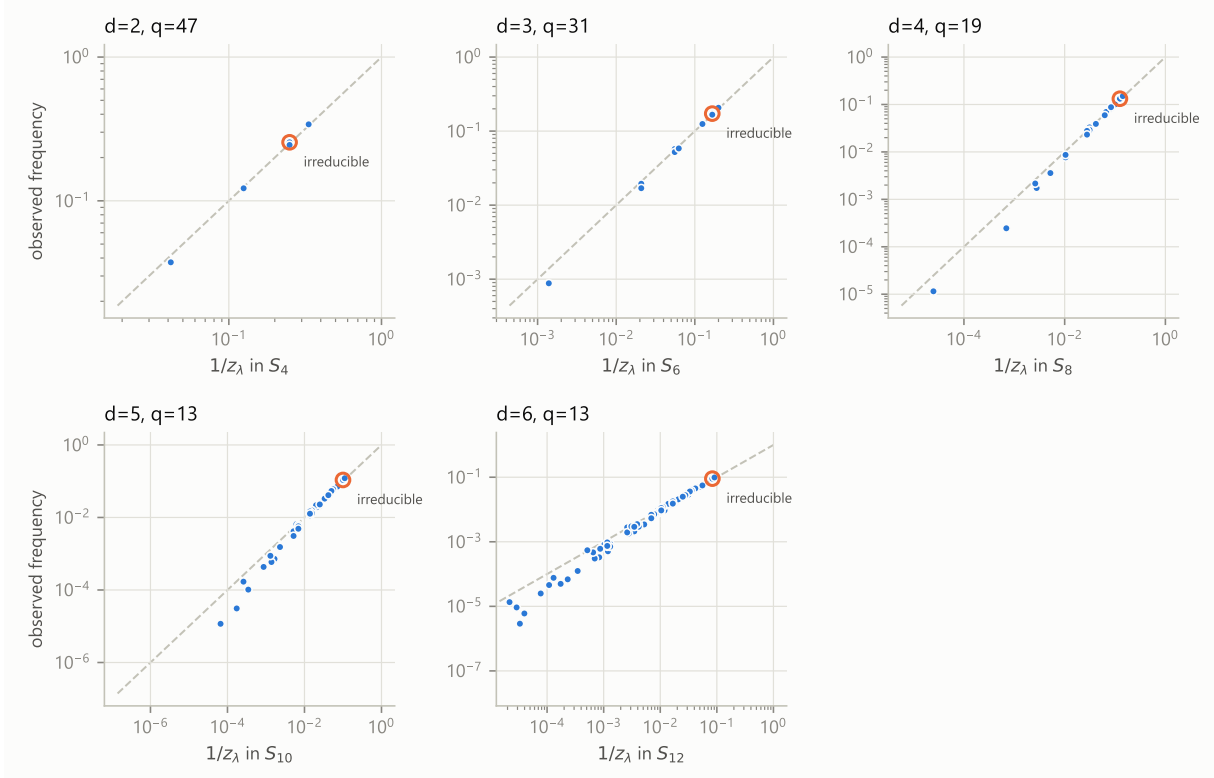


Figure 3: Observed frequency of each factorization type in  $\mathcal{I}_{t^d}$  against the  $S_{2d}$  cycle-type probability  $1/z_\lambda$ , as predicted by Theorem 5.7 and Chebotarev. The ringed point is the irreducible type.

- (4) Higher powers: the intervals  $\{f^k + s : \deg s \leq (k-1)d\}$  for  $k \geq 3$ , where the classical estimate covers a different range.

## A Errata to version v5

- E1. **Novelty.** The main theorem of v5,  $N_{\text{irr}} = q^{d+1}/(2d) + O_d(q^{d+1/2})$ , follows from the classical estimate of Theorem 1 and from [2, Thm. 2.3], without the squarefree hypothesis and in every odd characteristic.
- E2. **Primitivity.** v5 applied “imprimitive  $\Leftrightarrow F = g \circ h$ ” to the Galois group of  $F$  over  $\overline{\mathbb{F}}_q(\mathbf{a})$ , where it does not hold as stated, and treated algebraic coefficients as free parameters. Fixed in Proposition 5.5.
- E3. **Transposition.** v5 misidentified the discriminant branches near  $\mathbf{a} = 0$  and invoked Picard–Lefschetz without justification. Fixed in Proposition 5.6.
- E4. **Effective Chebotarev.** v5 stated the bound with the permutation sheaf, which detects linear factors, not  $2d$ -cycles, and cited an inapplicable theorem of [11].
- E5. **Betti-number bound.** v5 attributed to [12, Thm. 12] a bound for push-forwards of finite covers that is not there. The threshold  $(8d)^{2d+6}$  was unproved.
- E6. **Other.** An error-term mix-up in the final step, a  $j_1/j_*$  slip in the Grothendieck–Ogg–Shafarevich formula, wrong data for the Bary-Soroker citation, uncited references, version-

history remarks in the body, and conclusions drawn from three small examples.

## B Corrections to version v6

- F1. **Classical estimate omitted.** v6 did not mention the Hayes–Weil estimate (Theorem 1, [4, 10, 7]). It gives  $N_{\text{irr}} \geq 1$  for  $q \geq d - 1$  and error  $O(dq^d)$ . v6’s Theorem C ( $q > 1.65 \times 10^4$  for  $d = 4$ ), its Katz–Deligne bound and its bound via [19] (threshold about  $(2d + 2)^6$ ) are all superseded for existence; Table 2 compares them.
- F2. **Level 1/2.** v6 said that at level 1/2 the Riemann hypothesis gives an error larger than the main term “by a factor of order  $d$ ”. The factor is  $d/q$ , as Theorem 1 shows.
- F3. **Novelty of the twisted identity.** The root variety is Sawin’s  $X_{n,m,c}$ , so Theorem 6.2 is a twisted form of his point count (this was added late in v6).
- F4. **Scope.** v6 described  $q$  fixed,  $d \rightarrow \infty$  as open without locating the open range; it is  $q \leq d - 2$ .

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| $q$ | $d$             | intervals | orbits | $\min N_{\text{irr}}$ | ratio | $\text{Var } \Psi/q^{d+1}$ | $d - 2$ |
|-----|-----------------|-----------|--------|-----------------------|-------|----------------------------|---------|
| 2   | 2               | 1         | 1      | 1                     | 0.500 | 0.00                       | –       |
| 2   | 3               | 2         | 1      | 2                     | 0.750 | 0.00                       | –       |
| 2   | 4               | 2         | 2      | 2                     | 0.500 | 2.00                       | 2       |
| 2   | 5               | 4         | 2      | 5                     | 0.781 | 1.25                       | 3       |
| 2   | 6               | 4         | 3      | 4                     | 0.375 | 15.56                      | 4       |
| 2   | 7               | 8         | 4      | 13                    | 0.711 | 7.72                       | 5       |
| 2   | 8               | 8         | 6      | 28                    | 0.875 | 1.59                       | 6       |
| 2   | 9               | 16        | 8      | 47                    | 0.826 | 5.77                       | 7       |
| 2   | 10              | 16        | 10     | 92                    | 0.898 | 7.91                       | 8       |
| 2   | 11              | 32        | 16     | 167                   | 0.897 | 15.35                      | 9       |
| 2   | 12              | 32        | 20     | 296                   | 0.867 | 20.74                      | 10      |
| 2   | 13              | 64        | 32     | 593                   | 0.941 | 11.47                      | 11      |
| 2   | 14              | 64        | 36     | 1102                  | 0.942 | 17.37                      | 12      |
| 2   | 15              | 128       | 64     | 2106                  | 0.964 | 11.86                      | 13      |
| 2   | 16              | 128       | 72     | 3952                  | 0.965 | 15.22                      | 14      |
| 2   | 17              | 256       | 128    | 7580                  | 0.983 | 13.88                      | 15      |
| 2   | 18              | 256       | 136    | 14310                 | 0.983 | 19.71                      | 16      |
| 2   | 19              | 512       | 256    | 27290                 | 0.989 | 17.65                      | 17      |
| 2   | 20              | 512       | 272    | 51841                 | 0.989 | 19.45                      | 18      |
| 3   | 5               | 81        | 18     | 63                    | 0.864 | 1.80                       | 3       |
| 3   | 6               | 243       | 48     | 156                   | 0.856 | 4.85                       | 4       |
| 3   | 7               | 729       | 135    | 440                   | 0.939 | 4.44                       | 5       |
| 3   | 8               | 2187      | 378    | 1165                  | 0.947 | 5.50                       | 6       |
| 3   | 9               | 6561      | 1143   | 3123                  | 0.952 | 6.66                       | 7       |
| 3   | 10              | 19683     | 3321   | 8600                  | 0.971 | 7.89                       | 8       |
| 3   | 11 <sup>†</sup> | 59049     | 30     | 23996                 | 0.993 | –                          | 9       |
| 4   | 6               | 16        | 3      | 1280                  | 0.938 | 9.01                       | 4       |
| 4   | 7               | 64        | 8      | 4530                  | 0.968 | 9.36                       | 5       |
| 4   | 8               | 64        | 12     | 15936                 | 0.973 | 13.25                      | 6       |
| 4   | 9               | 256       | 24     | 57886                 | 0.994 | 6.36                       | 7       |
| 4   | 10              | 256       | 28     | 208536                | 0.994 | 10.47                      | 8       |
| 5   | 7               | 15625     | 815    | 27579                 | 0.988 | 4.45                       | 5       |
| 5   | 8               | 78125     | 3940   | 121334                | 0.994 | 5.51                       | 6       |

Table 1: The open range. “Intervals” is the number of distinct Legendre intervals and “orbits” the number examined after the affine reduction; rows marked <sup>†</sup> examine a random sample of orbits. The ratio is  $\min_f 2d N_{\text{irr}}(f)/q^{d+1}$ . The variance column is the mean of  $(\Psi(f) - q^{d+1})^2/q^{d+1}$  over all Legendre intervals. For odd  $q$  it is to be compared with the Keating–Rudnick limit  $d - 2$  [13], valid for  $d \geq 4$ ; for  $q = 2, 4$  the Legendre intervals are a special subfamily and no prediction applies. Rows with  $q \geq d - 1$  other than  $q = 2$  are omitted, since Theorem 1 covers them. Source: `data/open_regime_summary.csv`.

| $d$ | classical   | v5 (unproved) | Katz + Deligne | Cafure–Matera | via Sawin |
|-----|-------------|---------------|----------------|---------------|-----------|
| 1   | –           | 7.2           | 2.2            | 0.8           | –         |
| 2   | –           | 12.0          | 5.7            | 0.9           | 4.7       |
| 3   | <b>0.00</b> | 16.6          | 10.6           | 2.0           | 5.3       |
| 4   | <b>0.30</b> | 21.1          | 16.3           | 4.2           | 5.9       |
| 5   | <b>0.48</b> | 25.6          | 22.7           | 6.9           | 6.4       |
| 6   | <b>0.60</b> | 30.3          | 29.7           | 9.9           | 6.8       |
| 8   | <b>0.78</b> | 39.7          | 44.8           | 16.8          | 7.4       |
| 10  | <b>0.90</b> | 49.5          | 61.4           | 24.8          | 7.9       |
| 15  | <b>1.11</b> | 74.9          | 106.9          | 48.1          | 8.9       |
| 20  | <b>1.26</b> | 101.4         | 157.0          | 74.7          | 9.7       |
| 30  | <b>1.45</b> | 157.1         | 266.4          | 134.8         | 10.7      |
| 40  | <b>1.58</b> | 215.4         | 384.8          | 201.4         | 11.4      |
| 50  | <b>1.68</b> | 275.8         | 509.8          | 272.8         | 12.0      |

Classical:  $N_{\text{irr}} \geq 1$  for  $q > d - 2$  (Theorem 1); for  $d \leq 2$  for every  $q$ .

Table 2: Thresholds  $\log_{10} q_0(d)$  beyond which  $N_{\text{irr}} \geq 1$  is proved. “Classical” is Theorem 1 ( $q_0 = d - 2$ ). The other rigorous columns are the bounds of v6: Katz’s Betti bound with Deligne, Cafure–Matera, and Sawin’s theorem. “v5” was never proved.